

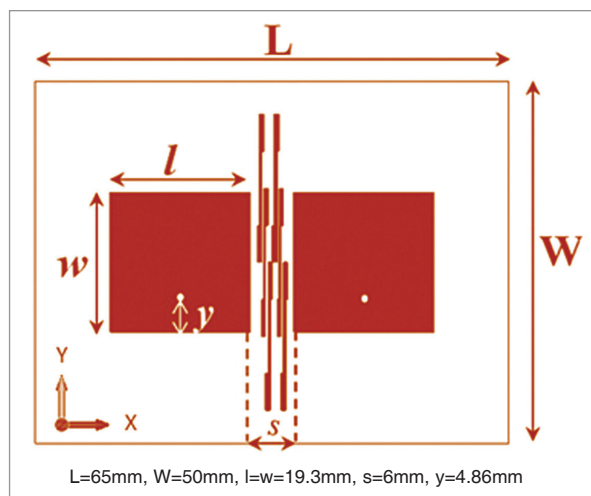


Fig. 3: Implementation of the decoupling element

coupled lines, slots, or stubs, ensure the electrical length and impedance are tuned so the structure presents a high impedance path at your operating frequency only.

- Check detuning: After adding the element, re-optimize feed positions or lengths, if needed, and always check that -10dB bandwidth and target centre frequency are preserved.

- Measure both S-parameters and patterns: Good S21 is not enough; verify that patterns, gain, and efficiency behave as expected.

- Think ahead to manufacturing: Avoid geometries that rely on extremely fine gaps or non-standard stackups, unless your production process can reliably achieve them.

Allocate a slim gap for the PCR

Here are some guidelines:

- Between the patches reserve a narrow strip (a few millimetres wide) for the PCR.

- Keep a small but non-zero spacing from patch edges to avoid detuning.

- Ensure there is enough length for the coupled line section

Check fabrication tolerances

Because the PCR's performance depends on small spacing and line widths:

- Use manufacturable dimensions (for example, avoid gaps below your PCB house's reliable limit).
- Keep line widths and spacing within standard tolerances to

reduce sensitivity.

- Consider a small parametric tolerance analysis in simulation to see how $\pm 0.05\text{--}0.1\text{mm}$ variations affect isolation.

Validate in the lab

On the bench:

- Calibrate the VNA carefully and use stable cables.
- Measure S11 and S21 for the array with and without the PCR.
- Use a simple anechoic-chamber or outdoor setup to capture patterns and gain at 3.5GHz .
- Compare measured results with simulation.

The example design used CST Microwave Studio with careful meshing, but the same workflow can be followed in any modern EM simulator. Often, a small, well-designed isolation structure plus minor feed adjustments is enough to rescue a compact layout that initially looked unusable.

The PCR-based mutual coupling reduction method is especially useful when:

- You must reduce element spacing to below 0.5λ in a 3.5GHz 5G design.

- Adding complex DGS or 3D structures is not feasible due to cost or mechanical restrictions.

- You want a planar, PCB-only solution that integrates into existing designs with minimal extra area.

Looking beyond 5G

As we move toward more advanced 5G deployments and future 6G concepts, the trend is toward even denser arrays, massive MIMO panels, multi-band modules, and integrated antenna-in-package solutions.

If you can realise a compact band-stop resonator using coupled lines at your frequency of interest, you can adapt the same approach to:

- Compact 5G base-stations or small-cell arrays.
- Customer-premises equipment and fixed wireless access terminals.
- Multi-band or multi-standard RF front-ends that must host several closely-spaced antennas.
- IoT gateways or industrial wireless nodes that have strict space constraints.

Whether the element is a parallel coupled-line resonator, a self-isolated radiator geometry, or a carefully shaped ground slot, the principle is the same: engineer the near-field environment so antennas can coexist more closely without 'talking' to each other. From my experience, designers who treat the gap as a design opportunity rather than a constraint are the ones who manage to ship denser, more capable 5G hardware without unpleasant surprises late in the cycle. **EFY**

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